MEDICATION SAFETY SIGNAL MONITOR · HAND-IN R07

Data judgment

Version 1.2, 6 November 2026: the problems we found in public drug-safety data, what we fixed, and what we left

The question this report answers

Which problems did we find in the public data, which did we fix and how, which did we leave, and why?

One-line answer

We logged 42 data issues. The ones that would put a wrong number or a wrong label in front of the safety officer are fixed by written rules with before/after evidence. The ones that only add noise are tagged rather than hidden. The ones we could not fix (no MedDRA licence, no pharmacist yet, only two FAERS quarters) are stated plainly below.

What this is	What this is not
• An honest account of data problems, grouped by what they would do to the officer's morning • Before/after numbers for each fix, with the rule (decision) that made it • A list of what we left and why, and what another month would change	• A complete clinical review: no pharmacist has checked the synonym bands, the AI answer key or the ranked signals yet • A claim that the ranked signals are real side effects: they are a review queue • The pipeline description (see R02) or a Foundry primer (see R01)

How to read this. Chapter 1 is the summary, with what it means for the officer. Chapters 2 to 6 group the problems by their effect on the officer: wrong counts, wrong label, noise in the queue, silent breaks, and recalls. Chapter 7 is what we left. Chapter 8 is what we would do with more time. Appendix A lists all 42 register entries. DQ numbers refer to `docs/07-data-quality/data-issues-register.md`; D numbers to `docs/06-decisions/DECISIONS.md`. Labels: OBSERVED measured, DERIVED computed from measured numbers, DESIGNED a choice we made (it may be built), ASSUMED not verified, TESTED built, run and checked.

Contents	Page
01 · Data judgment in one page	2
02 · Problems that change the counts	3
03 · Problems that give the wrong label	4
04 · Problems that fill the queue with noise	5
05 · Problems that break silently	6
06 · Recall problems	7
07 · What we left, and why	8
08 · What we would do with more time	9
A · The register, DQ01 to DQ42	10

01

Data judgment in one page

The brief warned that the data is messier than its documentation says. It was. We profiled every source by hand before building (D012) and kept a register of every problem: **42 entries**, DQ01 to DQ42. We sorted them by one question: *what would this do to the safety officer's morning?* The pilot covers 13 drugs: 8 opioids (oxycodone, hydrocodone, morphine, fentanyl, hydromorphone, methadone, buprenorphine, tramadol) and 5 GLP-1 drugs (semaglutide, liraglutide, dulaglutide, tirzepatide, exenatide).

Effect on the officer	Examples	What we did	Status
Wrong counts: a drug looks riskier or safer than the reports say	Case versions and deletes (DQ36), substring names (DQ34), suspect roles (DQ35), one patient filed hundreds of times (DQ37)	Four written rules in clean FAERS: D022, D020, D021, D023 v2	Fixed
Wrong label: a reaction looks *not in label* because we read the wrong one	Tablet vs injection label (DQ33), openFDA label lag (DQ28), HTTP 200 with no label (DQ42), no reference label (exenatide)	Reference label per product; DailyMed as truth; no label → no claim (G4)	Fixed
Noise at the top of the queue	Homicide, child neglect; efficacy terms; label synonyms; a spelling bug	Keyword buckets (tag, don't rank); reviewed synonym table	Reduced
Silent breaks: a feed changes and nothing errors	Reporter type blank 34% → 72% (DQ38); openFDA search quirks; DailyMed rate limits	Row-count checks; 21 monitoring checks per build	Watched
Things we cannot fix in a pilot	No MedDRA groupings; reports are not incidence; stimulated reporting; indication terms; no pharmacist	Stated, labelled, planned	Left

The ranked review queue shows the effect of the noise work on identical data OBSERVED:

Step	Ranked signals	Change	What it removed
Step 9 (first full build): 13 drugs, exact text + AI label check	1,355		baseline
+ spelling fix + synonym table syn_v1 (D043)	1,210	−145	label wording the exact check missed
+ wider buckets D016 v3	1,115	−95	efficacy, product complaints, procedures
+ synonym table syn_v2	1,073	−42	more wording misses, top 12 per drug

In all 282 items left the ranked queue DERIVED. None was deleted: each is still a signal with a bucket, a caution or a *possibly in label* status, so the officer can still find it.

What the cleaning kept

The known positive survived every step: semaglutide × NAION (optic ischaemic neuropathy, a sudden loss of blood flow to the optic nerve), 394 reports, PRR 112.5, now rank 1 for semaglutide. None of the semaglutide labels in DailyMed mention it. It is a *known positive*, not a discovery: EMA PRAC (6 Jun 2025) and a WHO alert (27 Jun 2025) named it (DECISIONS learning 2026-11-05).

What this means for Dana (the safety officer)

New signals arrive only with a new FAERS quarter. FAERS is about 130 days behind (monitoring `faers_age_days`), so *what changed since yesterday* is about recalls and labels, not new signals.

The FAERS feed itself is changing. The FDA replaced FAERS with AEMS (its new adverse event system) in March 2026 (DQ20). We use the published quarterly files; how the future feed will arrive is **unverified**. The reporter-type field already went from 34% to 72% blank (DQ38), which we watch every build.

Only the project owner can submit a triage decision today. Giving the officer that permission is a production step.

The AI's quotes are on her screen, for two drugs. The AI reads 5 labels (oxycodone and semaglutide); page 1 shows what it quoted. For the other drugs the label check she sees rests on exact text and the reviewed synonym table (report R08).

Gate: Chapter 1: problems sorted by what they would do to the officer TESTED

Required evidence: Every fix has a rule (D number) and a before/after count on the same data.

Result: Met for the rows marked Fixed and Reduced. *Left* rows say why.

If it fails: A fix without numbers is moved to chapter 7 (left) until measured.

02

Problems that change the counts

FAERS (the FDA Adverse Event Reporting System) is a database of spontaneous reports. Its bulk quarterly files are the only way to filter by a drug's role in a report, but they are messy. Four problems would have changed every count; all four are fixed in `clean_faers.py`.

DQ36: phantom rows from case versions (fixed by D022)

A report (case) gets follow-up versions in later quarters, and a later version can change the drug list. Delete lists point at any past quarter. In the October prototype, simply appending two quarters gave **8.7% phantom rows** (422,872 vs 389,074) OBSERVED (register, 2026-10-20).

Rule D022: keep the highest case version per case id (newer quarter wins ties), then apply that quarter's deletes to all history. In Foundry, Q2 replaced 26,681 Q1 cases and deleted 655: 6.9% of Q1's 397,224 cases changed in one quarter DERIVED. Result: 792,346 current cases.

DQ34: substring drug names (fixed by D020)

Matching *morphine* as a substring also matched apomorphine and diamorphine (heroin): about 50% extra morphine rows in the prototype OBSERVED. **Rule D020:** exact RxNorm ingredient name after splitting combination names and stripping salt words; never substring; unmatched names kept for review.

DQ35 and DQ23: who is a suspect (fixed by D021)

A report lists every drug the patient took. Our drug is often only *concomitant* (taken at the same time, not suspected). But counting only the *primary* suspect drops real reports: for oxycodone it would miss 56% of cases (Foundry, 2026-11-04) OBSERVED. **Rule D021:** primary suspect, secondary suspect or interacting; each ingredient once per case.

DQ37 and the mass-polypharmacy cluster (fixed by D023 v2)

One patient can be filed by many companies under different case ids. A fold on age, sex, country and event date (D023 v1) removed most of it. Then the first ranked oxycodone list was led by pemphigus and rheumatic fever. Behind them: a Canadian cluster of **465 of 3,408** oxycodone cases listing a median of **122 drugs**, with ages that change between filings, so the person key misses them (A012) OBSERVED.

Rule D023 v2 DESIGNED: every report with 30 or more drugs leaves the main count but is kept as `a_mass_reports`. A signal that is at least half mass reports gets the caution *mostly mass reports* and is not ranked. We chose to set aside, not delete: the evidence stays visible on the signal record.

> **How we know**
>
> Counts equal the October prototype exactly; the incremental path, which applies the rules quarter by quarter, gives identical rows (EXCEPT ALL both ways = 0).
>
> Baseline gate: for 5 random ranked signals, the count was recounted from FAERS rows: 5/5 equal.

03

Problems that give the wrong label

A signal needs two things: more reports than expected, and *not found in the label*. If we read the wrong label, or no label, a reaction looks unlabeled when it is not. That is the most damaging false positive, because it looks like a finding.

Problem	What we saw	Rule
DQ33 wrong product's label	openFDA tagged injection-era semaglutide reports as the oral tablet and linked the tablet label; injection-site pain then looked unlabeled	Reference label per product (application), D024 / D027
DQ07 many labels per drug	Brands, generics, repackagers each have labels	One reference label per product = the application owner's (G3)
DQ28 stale label copy	openFDA's label copy lagged DailyMed by months (Ozempic)	DailyMed is the label source of truth (D019)
DQ19, DQ40 section codes	Old labels use separate Warnings/Precautions; sub-sections use a generic code	Six safety section codes; fold everything nested (D037)
DQ27 version numbers	Version numbers run from 1 to 3,540 and mean nothing across labels	Compare versions only within one label (setid)

DQ42: an HTTP 200 that is not a label (handled)

For two tramadol repackager labels, DailyMed answered **HTTP 200 with the text** `None` instead of XML OBSERVED. A 200 is not proof of a document. The parser crashed the label build. Now raw keeps the reply as received; clean keeps only text starting with <; the version stays listed with `has_xml = false`. The DailyMed reuse step re-tries those two on every build on purpose (2 downloads per build).

Exenatide: no reference label, so no claim (D039 G4)

Exenatide has reports but no product with the owner's label in DailyMed. Rather than compare it with some other label, the rule says **uncertain is shown as uncertain**, never as *not in label*. So signals exist for 12 of our 13 drugs OBSERVED, and a monitoring check makes sure no ranked signal lacks a reference label (it is 0).

> **How we know**
> Baseline gate trace: for each of 5 random ranked signals, the reference labels searched were listed (for example 19 labels, 76 sections for morphine) and the reaction term was in none of their safety sections (`baseline-gate.md`).

04

Problems that fill the queue with noise

With the counts and labels right, the ranked list was still not what a pharmacist would want to read first. These items are *real reports* and *really not in the label*, but most are not new side effects. We **tag** them instead of deleting them DESIGNED.

Noise	Example at the top of the list	Fix	Effect
Social and legal terms	Child neglect, Homicide, Imprisonment led the first oxycodone list (A012)	D016 v2: keyword buckets; only *clinical* is ranked	Moved off the top
Abuse, overdose, pregnancy	Dependence and overdose dominate opioid reports (DQ15); neonatal withdrawal (DQ26)	Tagged buckets, visible but not ranked (D016)	Tagged
Label wording (synonyms)	"Impaired gastric emptying" for all 4 GLP-1s; the labels say "delays gastric emptying" (A014)	D043: reviewed synonym table; medium match → *possibly in label*	−145 with the spelling fix
A spelling bug	British spellings `-noea`, `-rrhoea`, `-ised` were missing from the US-spelling list: 9 hits wrongly unlabeled	Fixed in `ontology_signals.py`	part of the above
Efficacy and product complaints	Weight loss poor, Hunger, Increased appetite; "Product …" complaints (A013)	D016 v3: wider efficacy, product-use and administrative buckets	−95
More wording misses	Gastrointestinal hypomotility; injection-site terms; QT prolongation wording	Synonym table syn_v2 from the top 12 per drug	−42

The synonym table now has 258 rows: 69 high (same idea → in label), 72 medium (close idea → *possibly in label*, 108 signals) and 117 rejected (kept, so the same lead is not argued twice). Every phrase was checked to appear in that drug's own label.

> **Limits of this fix**
> The bands were set by Claude, not yet by a pharmacist. Medium matches are never ranked, so an error there hides a signal from the top of the list but not from the screen.
>
> Keyword buckets are a stand-in for MedDRA groupings (the standard medical term hierarchy), which need a licence we do not have. They will leak: D016 v3 deliberately keeps *weight increased* and eating-behaviour terms as clinical.

How we know

Each step was measured on identical data, locally and then in Foundry, with equal results (A013, A014).

Bucket changes do not change a signal's `evaluation_id`, so decisions already made stay linked. Synonym changes do change it, by design: the label check is part of what the officer saw.

05

Problems that break silently

The worst data problems raise no error. A field empties, a search drops a term, a server answers politely with nothing. We found these by comparing counts with each source's own total, and we now watch for them every build.

Problem	What happened	How we catch it now
DQ38 reporter type	FAERS reports with no reporter type: 34.3% in 2026 Q1, 72.1% in Q2. Likely the FDA's switch from FAERS to AEMS (DQ20) **ASSUMED**	Monitoring check per quarter: **warn**. We show *unknown* and never guess the reporter
openFDA OR search	Without parentheses the first term was silently dropped: 11 oxycodone recalls instead of 25	Row count must equal `meta.results.total` and the profile
openFDA sparse fields	Searching only the harmonised `openfda.*` field found 47 Drugs@FDA applications instead of 141	Every search also uses the source's own ingredient field
DailyMed rate limit	A fast local pull got HTTP 429 for 744 of 1,321 labels; each became an error row, not a gap	Retry with back-off; XML reuse cuts XML downloads to 2 per build (the index is still pulled, O3)
Recall volume	Recalls reported for our drugs: 10 in the last 4 weeks vs 3 the 4 before	Drift check: **warn**; the rise is real (A017)

Note on DQ38: 34% is the Foundry value at the baseline gate (`monitoring_checks`). The register's first profile on 20 October said 35%; this report uses the Foundry value.

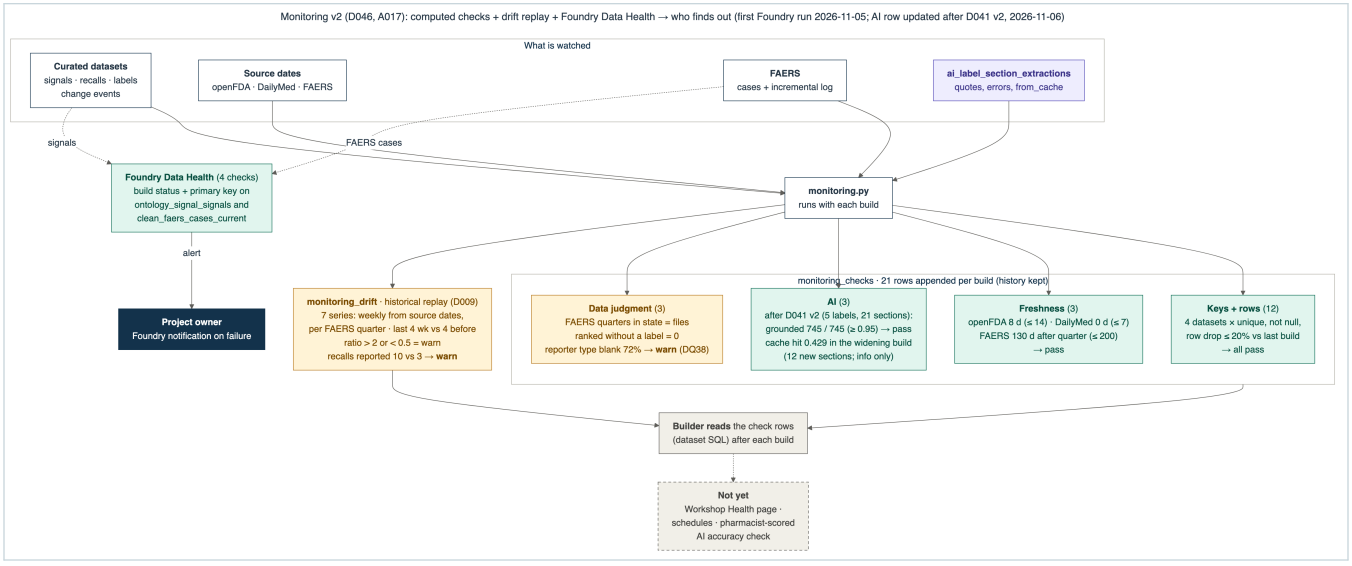


Figure 1. Monitoring (v2): 21 computed checks per build, the drift replay, and 4 Foundry Data Health checks, with first-run results (AI row updated after D041 v2).

Source: docs/diagrams/png/35-handin-monitoring-v2.png

How we know

Both warnings on the first Foundry run are real, which is the best evidence the checks work. The drift series is a **historical replay** from source dates, not weeks of live builds; live history accumulates with each build; there is no build schedule yet (D046).

06

Recall problems

Recalls (FDA enforcement reports) name products in free text. Linking a recall to *our* products is the hard part.

Problem	What we saw	What we did
Over-attached packages	openFDA attaches every package of an application to every recall record	Parse NDCs from the recall's own text; grade high / medium / low / none
DQ11, DQ29 NDCs missing or unknown	In the oxycodone pilot, 20 of 31 printed package NDCs were no longer in the NDC directory	Keep unmatched with a reason; every recall also gets an ingredient row
DQ12 compounded products	Newest recall D-0040-2027: compounded tirzepatide, no FDA product	Shown under tirzepatide; *Matched to our products: 0*, said plainly
DQ30 *Ongoing* for years	Some recalls stay Ongoing long after they matter	Active = Ongoing (126 of 900); sorted newest first; not tuned further

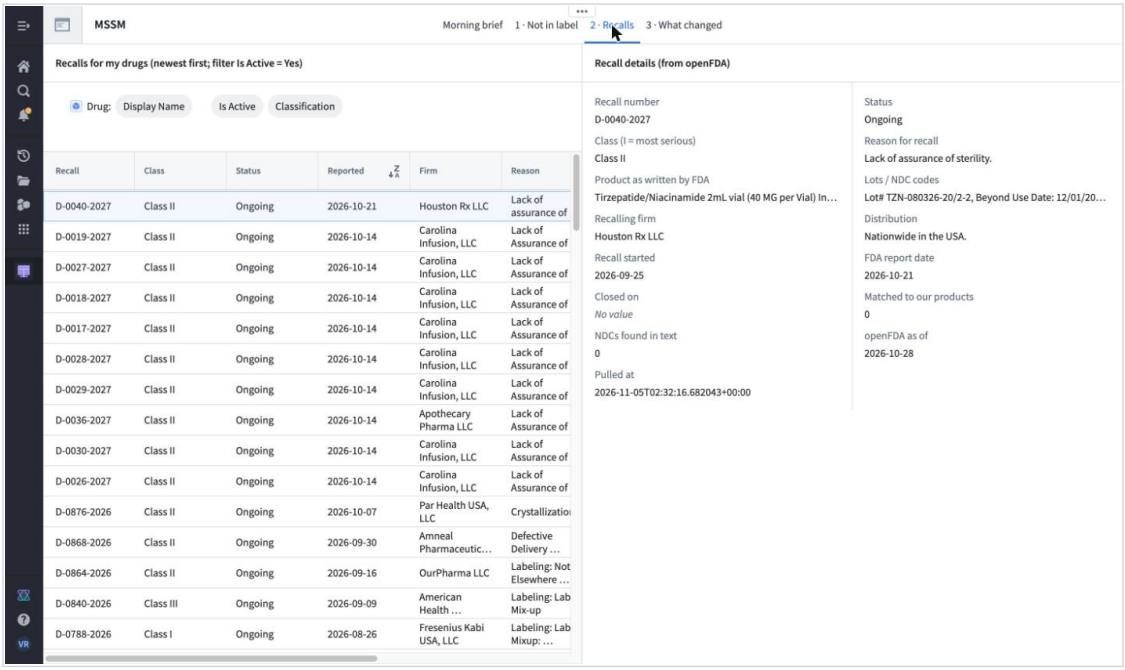


Figure 2. Page 2 · Recalls: active recalls for the chosen drugs, newest first; the detail panel shows D-0040-2027 with *Matched to our products: 0*.

Source: docs/11-handin/screens/app-2-recalls.jpg (Foundry, 2026-11-05)

07

What we left, and why

These are known and written down. Each one is a choice, not an oversight.

Left as is	Why	Risk to the officer
Reports are not incidence (DQ03)	FAERS has no denominator; PRR compares reporting, not risk	Wording on screen: *review queue*, not findings
Stimulated reporting (DQ04)	Spikes after news or lawsuits need event dates and more quarters than our two	A media-driven term can rank high. The hero signal NAION is the kind it can inflate; its reporter-type split is **not measured**
Indication terms (DQ06)	No rule yet to drop reactions that match the drug's own use (its Indications section)	Liraglutide's rank-1 signal is "Diabetic metabolic decompensation", a worsening of the diabetes it treats, not a new side effect
Lawyer-filed reports (DQ18, DQ21)	Tagging needs the reporter type, which went 72% blank (DQ38)	Litigation-style terms not separated
No MedDRA groupings (DQ22, DQ41)	Licence not available for a public pilot	One idea split over many terms (NAION spans about 10 optic terms); a grouped signal can be missed
Residue in the duplicate fold (DQ37)	The same patient with age 43 in one filing and 44 in another is not folded	Small double counts remain; the 30-drug cut catches the worst
Polydrug confounding (DQ17)	No adjustment model; role filter and mass-report cut only	A co-prescribed drug's effect can show up
AI long sections not scored (DQ10)	Answer key covers only Boxed Warning + Contraindications of 2 oxycodone labels (4 short sections); Claude-written, pharmacist pending. The 12 semaglutide sections are not scored	Unknown accuracy on Adverse Reactions tables; the AI does not change the list today
Synonym bands not reviewed	No pharmacist named yet	A wrong *high* band hides a real signal
Only two FAERS quarters (2026 Q1–Q2)	Pilot scope; enough for PRR, not for trends	No *new this quarter* trend yet

One modelling mistake, fixed

History, not data: a duplicate Signal → Drug link type created by mistake on 2026-11-05 was deleted on 6 Nov after checking nothing used it.

08

What we would do with more time

Next step	Fixes	Why first
A pharmacist review loop: Claude prepares an evidence packet; the pharmacist decides; the decision is saved by a create-only Foundry Action	Synonym bands (258 rows), the AI answer key, the top ranked signals	The biggest gap is not data but a named reviewer
MedDRA licence (or standard query groups)	DQ22, DQ41; replaces keyword buckets	Removes the largest source of leakage
More FAERS quarters	DQ04 stimulated reporting; *new this quarter* trends; re-measure incremental FAERS at 4+ quarters	Trends need history
Schedules and live drift	Freshness checks fire on their own; 3–4 week live drift window (D009)	Replay becomes real monitoring
Widen the AI to the other drugs' reference labels (today oxycodone and semaglutide)	DQ10; quotes for every *in label* decision	Only after the answer key is signed off
Score the long sections with a pharmacist	DQ10; the semaglutide and Adverse Reactions sections are unscored	Turns grounding into accuracy
Second opinions	SrLC label changes, FDA Drug Safety Communications, OnSIDES, PubMed and ClinicalTrials.gov in the packet	Researched in `docs/09-roadmap/another-month-sources-and-pharmacist-review.md`

Gate: Data judgment: found, fixed what matters, honest about the rest TESTED

Required evidence: A register of every issue; a rule and before/after numbers for each fix; a reason for each item left.

Result: Met. 42 issues logged; wrong-count and wrong-label problems fixed by rule; queue noise cut by 282 ranked items without deleting any; silent breaks watched; ten items left with reasons.

If it fails: Any new issue goes into the register first, with its effect on the officer.

A

The register, DQ01 to DQ42

The register's own status column was set when each issue was first logged; many rows still say *Suspected* although a later decision handled them. The right-hand column reads the decisions log.

ID	Issue	Register says	Where it stands
DQ01	FAERS follow-up versions duplicate cases	Suspected	Handled by D022 (see DQ36)
DQ02	Drug names are free text	Suspected	Handled: D020 exact identity, unmatched kept
DQ03	Reports are not incidence	Suspected	Accepted: PRR + careful wording
DQ04	Stimulated reporting after news	Suspected	Left (chapter 7)
DQ05	Suspect vs concomitant role	Suspected	Handled: D021
DQ06	Indication terms reported as reactions	Suspected	Left (known gap): indication terms still rank (liraglutide #1 "Diabetic metabolic decompensation")
DQ07	Many labels per ingredient	Suspected	Handled: reference label per product (G3)
DQ08	Label version history	Suspected	Handled: per setid + version; change feed
DQ09	Label prose vs FAERS terms	Handled partly	Text + US spelling + synonyms syn_v2; pharmacist pending
DQ10	Reaction tables inside label XML	Suspected	Left: long sections not scored
DQ11	Recall product in free text	Suspected	Handled: NDC from text + confidence
DQ12	Compounded products recalled	Suspected	Shown under the drug; 0 product matches
DQ13	Ingredient vs salt names	Suspected	Handled: salt strip (D020)
DQ14	Sources on different clocks	Suspected	Handled: as-of per source (D045)
DQ15	Abuse/overdose dominate opioid reports	Confirmed	Tagged, not ranked (D016)
DQ16	Combination products	Suspected	Flagged; each ingredient counted
DQ17	Polydrug confounding	Suspected	Left: role filter + mass cut only
DQ18	Litigation-style terms	Confirmed	Left: reporter type unreliable (DQ38)
DQ19	Old vs new Warnings section codes	Confirmed	Handled: both codes (D037)
DQ20	FAERS replaced by AEMS (Mar 2026)	Confirmed	Quarterly files used; future feed unverified
DQ21	Lawyers filed most 2021 oxycodone reports	Confirmed	Left: D025 monitors fill rate
DQ22	"Hyperalgesia" has 0 reports	Confirmed	Left: needs MedDRA grouping
DQ23	Our drug often only concomitant	Confirmed	Handled: bulk files + D021
DQ24	openFDA enrichment is a name match	Confirmed	Avoided: bulk files + own identity
DQ25	Search field changes counts	Confirmed	Handled: one identity rule
DQ26	Pregnancy/neonatal terms rank high	Confirmed	Tagged, not ranked (D016)

ID	Issue	Register says	Where it stands
DQ27	Version numbers not comparable	Confirmed	Compared within one setid only
DQ28	openFDA label copy lags DailyMed	Confirmed	Handled: DailyMed is truth (D019)
DQ29	Compounded NDCs unknown	Confirmed	Kept with a reason; ingredient row
DQ30	Recalls Ongoing for years	Confirmed	Active = Ongoing; newest first
DQ31	ATC class misses our drugs	Confirmed	Handled: explicit 13-RxCUI list (D018)
DQ32	Trials data noisy for safety	Confirmed	Source cut from the pilot (D031)
DQ33	Tablet vs injection label	Confirmed	Handled: product-level label (D024; D027)
DQ34	Substring drug matching	Measured	Handled: D020
DQ35	SS role repeats PS	Measured	Handled: D021, once per case
DQ36	Case versions and deletes: phantom rows	Measured	Handled: D022
DQ37	Same patient under many case ids	Measured	Handled: D023 v1 + v2; residue left
DQ38	Reporter type blank jumps	Measured	Watched: monitoring warn
DQ39	Bulk FAERS has no openfda block	Measured	Handled: `nda_num` → application (D024)
DQ40	Sub-section codes; repeated version	Measured	Handled: fold nested; text hash in cache key
DQ41	One idea over many terms (NAION)	Measured	Left: no MedDRA grouping
DQ42	HTTP 200 with "None"	Handled	Handled: clean keeps only XML

Figures quoted from project records (not in numbers.json)

Figure	Source
Append = 8.7% phantom rows (422,872 vs 389,074)	DQ36, register (prototype, 2026-10-20)
Morphine substring about +50% rows	DQ34, register (prototype)
Primary-suspect-only misses 56% of oxycodone cases	DECISIONS learnings, 2026-11-04
465 of 3,408 oxycodone cases, median 122 drugs	D023 v2; A012
Ranked removals by bucket; gastric emptying; 9 spelling hits	A013, A014
openFDA 11 vs 25 recalls; 47 vs 141 applications	DECISIONS learnings, 2026-11-04
DailyMed 429 for 744 of 1,321 labels (local pull)	DECISIONS learnings, 2026-11-04
20 of 31 printed recall NDCs not in the directory	DECISIONS learnings, 2026-11-04
Version numbers 1 to 3,540	DQ27, register
Morphine trace: 19 labels, 76 sections	baseline-gate.md
Liraglutide rank-1 "Diabetic metabolic decompensation"	DQ06; Workshop page 1 (screen app-1, 2026-11-05)
Reporter type blank 35% (first profile)	DQ38, register (profile 2026-10-20)

Figure	Source
AEMS replaced FAERS in March 2026	DQ20, register
About 30 DailyMed index pages still pulled per build	O3, optimization log

Numbers used in this report

Every number in the table below is looked up from `docs/11-handin/numbers.json` at build time. Source: Foundry SQL (repo SQL panel) after the baseline-gate rebuild of the whole downstream chain: raw_dailymed_spl → clean_labels → clean_label_product_map → ontology_objects → ai_label_extraction → ontology_signals → ontology_changes → monitoring. Build durations from the Foundry Builds page. Query: docs/11-handin/baseline-gate.md. Captured: 2026-11-05T07:50Z. Figures quoted from other project records (decision log, data-issues register, optimization log) name that record where they appear.

Key	Value
`ai.d041_v2.new_sections`	12
`ai.pilot_labels`	5
`ai.v1_pilot.labels`	2
`faers.cases_current`	792,346
`faers.q1_cases`	397,224
`faers.q2_cases_deleted`	655
`faers.q2_cases_replaced`	26,681
`monitoring.checks_per_build`	21
`monitoring.data_health_checks`	4
`monitoring.faers_age_days`	130
`monitoring.recalls_last4w`	10
`monitoring.recalls_prev4w`	3
`monitoring.reporter_type_blank_q1`	0.343
`monitoring.reporter_type_blank_q2`	0.721
`optimization.O3.after_downloads`	2
`recalls.active`	126
`recalls.newest`	D-0040-2027
`recalls.total`	900
`scope.drugs_formulary`	13
`scope.drugs_with_signals`	12
`scope.glp1`	5
`scope.opioids`	8

Key	Value
signals.naion.a	394
signals.naion.prr	112.5
signals.naion.rank	1
signals.possibly_in_label	108
signals.ranked_history.d016_v3	1,115
signals.ranked_history.step9	1,355
signals.ranked_history.syn_v1	1,210
signals.ranked_history.syn_v2	1,073
synonyms.high	69
synonyms.medium	72
synonyms.rejected	117
synonyms.rows	258